HEART DISEASE RISK ANALYSIS AND PREDICTION USING DEEP LEARNING

A COMPREHENSIVE CASE STUDY IN EXPLORATORY DATA ANALYSIS, STATISTICAL INFERENCE, AND NEURAL NETWORK ARCHITECTURES

A Case Study Technical Report submitted in fulfillment of the course assessment for
U21ADPO5 - EXPLORATORY DATA ANALYSIS AND VISUALIZATION

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GitHub Repository: https://github.com/srenishh18/heart-disease-risk-analysis-using-deep-learning

Dataset Source: https://www.kaggle.com/datasets/dileep070/heart-disease-prediction-using-logistic-regression



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ABSTRACT

Cardiovascular disease remains the leading cause of mortality globally, making early and accurate prediction of coronary heart disease (CHD) risk a critical public health priority. This case study presents an end-to-end predictive analytics pipeline applied to the Framingham Heart Study benchmark dataset, comprising 4,240 longitudinal patient records and 15 demographic, behavioral, and clinical risk factor attributes. Through systematic Exploratory Data Analysis (EDA), distribution-aware preprocessing, and a TensorFlow/Keras Multi-Layer Perceptron (MLP) deep learning model, this study investigates the predictive drivers of 10-year CHD risk.

The preprocessing pipeline addresses missing value imputation (388 missing glucose readings handled via median strategy), class imbalance (85% negative / 15% positive), and feature normalization through StandardScaler. Six exploratory and four diagnostic visualizations are generated in adherence to Ben Fry's seven stages of data visualization. The trained MLP model, with an architecture of Dense(64)–BatchNorm–Dropout(0.30)–Dense(32)–Dropout(0.20)–Dense(1, sigmoid), achieves a **Test Accuracy of 85.38%** and a **Test ROC-AUC of 0.6873** on the 636-sample hold-out test partition, converging via Early Stopping at Epoch 16 out of a maximum 40 epochs.

Keywords: Exploratory Data Analysis, Heart Disease Prediction, Coronary Heart Disease, Framingham Heart Study, TensorFlow, Multi-Layer Perceptron, ROC-AUC, Deep Learning.

TABLE OF CONTENTS & ROADMAP

1. INTRODUCTION & OBJECTIVES 5
1.1 Context and Operational Motivation 5
1.2 Problem Statement & Mathematical Formulation 5
1.3 Project Objectives (CO3, CO4, CO5) 0
1.4 Summary of Technical Contributions 5
1.5 Report Organization & Roadmap 6
1.6 Literature Review (5 Sources) 0
2. DATASET DESCRIPTION & FEATURE ANALYSIS 6
2.1 Provenance, Scope, and Data Source Integrity 6
2.2 Dataset Dimensions & Memory Footprint 6
2.3 Raw Feature Dictionary (Table 1) 0
2.4 Summary Statistics 7
2.5 Target Variable Characteristics (TenYearCHD) 0
2.6 Missing Value Audit 7
3. EXPLORATORY DATA ANALYSIS & PREPROCESSING 8
3.1 Missing Value Mechanism & Imputation Protocols 8
3.2 Outlier Detection & Handling Strategies 8
3.3 Deduplication & Sanity Validations 8
3.4 Class Imbalance Analysis 8
3.5 Encodings & Normalization 9
3.6 Dataset Partitioning & Stratification 9
4. COMPREHENSIVE DATA VISUALIZATION CATALOG 9
4.1 Ben Fry's Seven Stages Mapping (Table 2) 0
4.2 Suitability Rationale for Chart Typologies (Table 3) 0
4.3 Exploratory Visualization Catalog (Figs. V01–V06) 0
4.4 Bivariate and Multivariate Synthesis 0
5. DEEP LEARNING MODEL & BASELINES 14
5.1 MLP Architecture Specification (Table 4) 0
5.2 Training Regimen & Optimization Hyperparameters 14
5.3 Preprocessing Pipeline Design 15
6. RESULTS & PERFORMANCE INTERPRETATION 15
6.1 Multi-Metric Performance Benchmark (Table 5) 0
6.2 Diagnostic Visualizations (Figs. E01–E04) 0
6.3 Performance Diagnosis: Class Imbalance Effects 18
7. CONCLUSION, RECOMMENDATIONS & FUTURE SCOPE 19
7.1 Synthesis of Empirical Learnings 19
7.2 Model Performance vs. Baseline Benchmark 19
7.3 Limitations 19
7.4 Future Research Roadmap 19
8. REFERENCES (IEEE FORMAT) 20
9. APPENDIX: CODEBASE & REPOSITORY STRUCTURE 20

LIST OF FIGURES

Fig. V01. Ten-Year CHD Outcome Distribution (Target Count Plot) 10
Fig. V02. Distribution of male (Histogram + KDE) 10
Fig. V03. Outlier Check for male (Box Plot) 11
Fig. V04. Correlation Heatmap of Numerical Features 12
Fig. V05. male by CHD Outcome (Bivariate Box Plot) 13
Fig. V06. Age by Ten-Year CHD Outcome (Bivariate Box Plot) 13
Fig. E01. Training vs. Validation Loss (Loss vs. Epoch) 16
Fig. E02. Training vs. Validation Accuracy (Accuracy vs. Epoch) 16
Fig. E03. Confusion Matrix 17
Fig. E04. ROC Curve (MLP, AUC = 0.687) 18

LIST OF TABLES

Table 1. Raw Feature Dictionary (15 Input Fields + Target) 7
Table 2. Ben Fry's Seven Stages Mapping to Study Artifacts 9
Table 3. Suitability Rationale for Selected Chart Typologies 9
Table 4. MLP Architectural Specifications 14
Table 5. Comprehensive Model Performance Benchmark 15

1. INTRODUCTION & OBJECTIVES

1.1 Context and Operational Motivation

Cardiovascular diseases (CVDs) account for approximately 17.9 million deaths annually, representing 31% of all global fatalities (World Health Organization, 2021). Among CVDs, coronary heart disease (CHD) — characterized by the narrowing or blockage of coronary arteries — is the most prevalent and deadly form. The ability to identify high-risk individuals years before a cardiac event enables physicians to implement preventive interventions: lifestyle modifications, pharmacological management of hypertension and cholesterol, and early surgical assessment.

The Framingham Heart Study, initiated in 1948, remains the most historically significant longitudinal cohort study in cardiovascular epidemiology [6]. By tracking thousands of Framingham, Massachusetts residents over decades, it identified fundamental risk factors — systolic blood pressure, serum cholesterol, smoking, diabetes, and age — that now underpin modern cardiovascular risk assessment frameworks including the widely adopted Framingham Risk Score. In the machine learning era, this dataset serves as a benchmark for evaluating the clinical utility of predictive models in preventing premature CHD events.

1.2 Problem Statement & Mathematical Formulation

This study addresses a supervised binary classification task under strict data integrity constraints. Let $D = \{(x_i, y_i)\}$, $i = 1 \ldots N$, represent the dataset of $N = 4{,}240$ patient records, where x_i is a 15-dimensional clinical feature vector ($x_i \in R^{15}$) observable at baseline examination, and $y_i \in \{0, 1\}$ indicates whether the patient developed CHD within 10 years ($y = 1$) or remained disease-free ($y = 0$). The objective is to learn a predictive mapping

$$f : R^{15} \rightarrow [0, 1]$$

that estimates the posterior probability $P(y = 1 \mid x)$, optimized under Binary Cross-Entropy (BCE) loss:

$$BCE = -(1/N) \Sigma [\, y_i \cdot \log f(x_i) + (1 - y_i) \cdot \log(1 - f(x_i)) \,]$$

Classification is achieved by thresholding posterior probabilities at 0.5. Performance is evaluated via Test Accuracy, ROC-AUC, and class-specific Precision, Recall, and F1-Score metrics to account for the inherent class imbalance.

1.3 Project Objectives

This project addresses the curricular competencies of Course Outcomes CO3, CO4, and CO5:

- **CO3 (Data Visualization – 8 Marks):** Design, implement, and interpret an exhaustive catalog of exploratory visualizations adhering to Ben Fry's seven stages of data visualization (Acquire, Parse, Filter, Mine, Represent, Refine, Interact).
- **CO4 (Statistical & Distribution-Based Visualization – 16 Marks):** Conduct rigorous distributional analysis on continuous risk factors, evaluate missing value mechanisms, analyze class imbalance, and visualize correlation structures and bivariate relationships.
- **CO5 (Application of Data Exploration & Visualization – 16 Marks):** Architect, train, and evaluate a TensorFlow/Keras Multi-Layer Perceptron, diagnose failure modes via confusion matrix and ROC analysis, and interpret training dynamics via loss and accuracy learning curves.

1.4 Summary of Technical Contributions

1. **End-to-End Pipeline:** Complete EDA → preprocessing → deep learning → evaluation pipeline on the Framingham dataset with reproducibility seed = 42.
2. **Comprehensive Missing Value Handling:** Identified and imputed missing values in 7 columns (totalling 645 missing cells), applying median imputation via SimpleImputer under strict train-only fit constraints.
3. **Six Exploratory Visualizations:** Target distribution, feature distribution, outlier box plot, correlation heatmap, and bivariate relationship plots.

4. **MLP Deep Learning Model:** Two-hidden-layer neural network (64→32→1) with BatchNormalization and Dropout regularization, achieving 85.38% test accuracy.
5. **Rigorous Evaluation:** Full evaluation suite including classification report, confusion matrix, ROC curve (AUC = 0.687), and learning curves.

1.5 Report Organization & Roadmap

Section 2 characterizes the Framingham dataset schema, data source integrity, summary statistics, and feature space. Section 3 details EDA, missingness handling, outlier strategies, and class imbalance. Section 4 presents the comprehensive visualization catalog. Section 5 specifies the deep learning MLP architecture and training regimen. Section 6 provides benchmark results and diagnostic visualizations. Section 7 concludes with recommendations and future research directions. Section 8 provides IEEE-formatted references, and Section 9 contains the technical appendix.

1.6 Literature Review

Exploratory Data Analysis was systematically formalized by John Tukey [1], who emphasized the primacy of visual pattern discovery over confirmatory statistical testing. Tukey's resistant summary statistics — medians, hinges, and box plots — remain foundational tools for identifying distributional asymmetries and outliers in clinical data. Ben Fry [2] later codified computational visualization into seven sequential stages (Acquire, Parse, Filter, Mine, Represent, Refine, Interact), providing a structured methodology for translating raw tabular data into interpretable graphical insights.

In medical predictive modeling, Pedregosa et al. [3] demonstrated in Scikit-Learn the practicality of modular pipelines that integrate imputation, scaling, and model fitting under strict leakage controls. In deep learning for clinical tabular data, Goodfellow et al. [4] established theoretical foundations for multi-layer perceptrons, batch normalization, and dropout regularization — all employed in this study. Addressing class imbalance in clinical datasets, Chawla et al. [5] demonstrated that standard models trained on skewed distributions systematically under-detect minority class events, motivating performance evaluation beyond accuracy alone (specifically, ROC-AUC and per-class F1-Score).

2. DATASET DESCRIPTION & FEATURE ANALYSIS

2.1 Provenance, Scope, and Data Source Integrity

The dataset analyzed in this case study is the **Framingham Heart Study dataset**, publicly accessible via Kaggle. The data originates from the landmark National Heart, Lung, and Blood Institute (NHLBI) longitudinal cardiovascular epidemiology study initiated in 1948, tracking residents of Framingham, Massachusetts. The version used here is a curated, anonymized tabular benchmark reflecting baseline examination measurements. The dataset constitutes a single cohort file (framingham.csv) with no separate test partition provided; a stratified 70/15/15 train/validation/test split was implemented internally under reproducibility seed 42.

2.2 Dataset Dimensions & Memory Footprint

The raw dataset comprises **4,240 patient records** across **16 columns** (15 clinical/demographic input features + 1 binary target column TenYearCHD), occupying approximately 192 KB on disk. After removing 0 duplicate rows and dropping rows with missing target values, the cleaned dataset retains all 4,240 records. The final modeling dataset uses 15 input features after removing the target column.

2.3 Raw Feature Dictionary

Table 1. Raw Feature Dictionary (15 Input Fields + Target)

Feature	Data Type	Description	Missing %
male	Binary (int)	Sex: 1 = Male, 0 = Female	0.0%
age	Integer	Age of patient in years	0.0%
education	Categorical (float)	Education level: 1 (some HS) to 4 (college+)	2.48%
currentSmoker	Binary (int)	Currently smoking: 1 = Yes, 0 = No	0.0%
cigsPerDay	Continuous (float)	Cigarettes smoked per day	0.68%
BPMeds	Binary (float)	On blood pressure medication: 1 = Yes	1.25%
prevalentStroke	Binary (int)	Prior stroke history: 1 = Yes	0.0%
prevalentHyp	Binary (int)	Hypertensive: 1 = Yes	0.0%
diabetes	Binary (int)	Diabetic: 1 = Yes	0.0%
totChol	Continuous (float)	Total serum cholesterol (mg/dL)	1.18%
sysBP	Continuous (float)	Systolic blood pressure (mmHg)	0.0%
diaBP	Continuous (float)	Diastolic blood pressure (mmHg)	0.0%
BMI	Continuous (float)	Body Mass Index (kg/mZ)	0.45%
heartRate	Continuous (float)	Resting heart rate (beats/min)	0.02%
glucose	Continuous (float)	Fasting blood glucose (mg/dL)	9.15%
TenYearCHD	**Binary (int)**	**Target: 10-year CHD onset (1 = Yes, 0 = No)**	**0.0%**

2.4 Summary Statistics

Key numerical summary statistics for the training cohort (n = 2,968):

Feature	Mean	Std	Min	Q1	Median	Q3	Max
age	49.6	8.6	32	42	49	56	70
cigsPerDay	9.0	11.9	0	0	0	20	70
totChol	236.7	44.6	113	206	234	263	696
sysBP	132.4	22.1	83.5	117.5	128	143.9	295
diaBP	82.9	12.0	48	75	82	90	142.5
BMI	25.8	4.1	15.5	23.1	25.4	28.0	56.8
heartRate	75.7	12.0	44	67	75	83	143
glucose	81.9	23.5	40	71	78	87	394

2.5 Target Variable Characteristics

The ground-truth response variable TenYearCHD is a binary indicator of 10-year coronary heart disease onset. In the full cleaned dataset:

- **Class 0 (No CHD):** 3,596 patients — **84.8%**
- **Class 1 (CHD):** 644 patients — **15.2%**

This pronounced **class imbalance ratio of approximately 5.6:1** is a fundamental characteristic of the dataset and has direct implications for model evaluation: raw accuracy alone is misleading (a naïve majority classifier achieves 84.8%), necessitating ROC-AUC and per-class recall as primary performance metrics.

2.6 Missing Value Audit

A systematic audit revealed missing values concentrated in 7 input columns (645 missing cells in total):

Column	Missing Count	Missing %	Imputation Strategy
glucose	388	9.15%	Median imputation
education	105	2.48%	Median imputation
BPMeds	53	1.25%	Median imputation
totChol	50	1.18%	Median imputation
cigsPerDay	29	0.68%	Median imputation
BMI	19	0.45%	Median imputation
heartRate	1	0.02%	Median imputation

All remaining 8 input columns exhibit 100% completeness, and the TenYearCHD target column has 0 missing values. Because every feature is numerically encoded in the dataset, all columns pass through the numeric (median) imputation branch of the preprocessing pipeline.

3. EXPLORATORY DATA ANALYSIS & PREPROCESSING

3.1 Missing Value Mechanism & Imputation Protocols

To determine whether missingness is informative with respect to the target, the proportion of CHD-positive cases was compared between missing and non-missing subsets for the most critical column (glucose): missing glucose records exhibited a CHD rate of 14.4% vs. 15.3% for observed records, indicating that the missingness mechanism is approximately Missing Completely At Random (MCAR) and non-informative.

Imputation was executed under strict leakage controls: all imputation parameters (medians) were computed exclusively on the training partition using Scikit-Learn SimpleImputer, then applied identically to the validation and test sets via the preprocessing pipeline's fit_transform / transform separation.

3.2 Outlier Detection & Handling Strategies

Numerical features were examined for extreme values using Tukey's IQR rule (Q3 + 1.5 × IQR):

- **totChol:** Maximum of 696 mg/dL is clinically extreme (IQR upper fence ≈ 347 mg/dL), identifying approximately 0.8% of records as potential outliers.
- **glucose:** Maximum of 394 mg/dL indicates severe hyperglycemia; IQR fence ≈ 111 mg/dL.
- **sysBP:** Maximum of 295 mmHg (hypertensive crisis range); 1.2% of records flagged.

Following the same rationale as clinical epidemiological practice, **all outlier records are retained**. In longitudinal cardiac studies, extreme values in cholesterol, blood pressure, and glucose are medically valid data points representing the high-risk population most relevant to CHD prediction. Removing them would systematically bias model performance toward low-risk patients. The MLP model's use of StandardScaler normalization mitigates the influence of extreme values on gradient propagation.

3.3 Deduplication & Sanity Validations

Data integrity audits confirmed **zero duplicate rows** (df.duplicated().sum() = 0). Domain boundary validations were performed: age ranges from 32–70 years (valid adult cohort), BMI ranges from 15.5–56.8 (clinically plausible), and heart rate ranges from 44–143 bpm (physiologically valid). No corrupt or unphysical records were detected.

3.4 Class Imbalance Analysis

The 5.6:1 negative-to-positive class ratio was analyzed as the primary challenge in this dataset. A naïve majority classifier would achieve 84.8% accuracy while detecting 0% of actual CHD cases. The trained MLP model, evaluated on the test set, achieves 99% recall on Class 0 but only 7% recall on Class 1 — confirming that the model learns the dominant majority class pattern. Mitigation strategies (class weighting, SMOTE oversampling) are identified as key future improvements.

3.5 Encodings & Normalization

All 15 input features are numerically encoded in the source file (binary indicators are stored as 0/1 and education as an ordinal 1–4 level). The pipeline therefore routes every feature through the numeric branch: **median imputation → StandardScaler normalization**. A categorical branch (most-frequent imputation → OneHotEncoder) is defined for robustness but receives an empty column list for this dataset. The ColumnTransformer outputs a unified float32 feature matrix of dimension N × 15.

3.6 Dataset Partitioning & Stratification

The cleaned dataset (N = 4,240) is split into three stratified partitions preserving the 85/15 class ratio:

Partition	Samples	CHD Positive	CHD Negative
Training (70%)	2,968	451 (15.2%)	2,517 (84.8%)
Validation (15%)	636	97 (15.3%)	539 (84.7%)
Test (15%)	636	96 (15.1%)	540 (84.9%)

Stratification was enforced via train_test_split(stratify=y) to maintain representative class distributions across all three partitions.

4. COMPREHENSIVE DATA VISUALIZATION CATALOG

4.1 Ben Fry's Seven Stages of Data Visualization

Table 2. Ben Fry's Seven Stages Mapping to Study Artifacts

Stage	Notebook Section	Figure ID	Primary Artifacts
1. Acquire	Data Loading	—	framingham.csv (4,240 × 16)
2. Parse	Schema Audit	—	df.dtypes, df.info() output
3. Filter	Missing & Dedup	V03	Outlier box plot, missing value table
4. Mine	Distribution Analysis	V01, V02	Target countplot, feature histogram
5. Represent	Visualization	V01–V06	All 6 EDA figures
6. Refine	Polished Charts	V04	Correlation heatmap (coolwarm, centered at 0)
7. Interact	Evaluation	E01–E04	Learning curves, confusion matrix, ROC

4.2 Suitability Rationale for Chart Typologies

Chart types were selected according to the data type and analytical question of each figure; all charts were produced with Matplotlib and Seaborn [7].

Table 3. Suitability Rationale for Selected Chart Typologies

Typology	Figure	Suitability Rationale
Count Plot (Bar)	V01	Directly visualizes class imbalance — absolute counts communicate the 3,596 vs. 644 disproportion effectively
Histogram + KDE	V02	Reveals the distribution shape, skewness, and modality of a feature without bin-width artifacts
Box Plot (Univariate)	V03	Compactly displays Q1/Q3/IQR and individual outlier points beyond 1.5×IQR fences
Heatmap (Correlation)	V04	Enables rapid visual scanning of pairwise collinearity among all 15 numerical features simultaneously
Box Plot (Bivariate)	V05, V06	Contrasts feature distribution shifts between CHD+ and CHD− groups, communicating median differences and overlap

4.3 Exploratory Visualization Catalog

Each figure is accompanied by three structured blocks: *What it shows*, *Why it was created*, and *Insights obtained.*

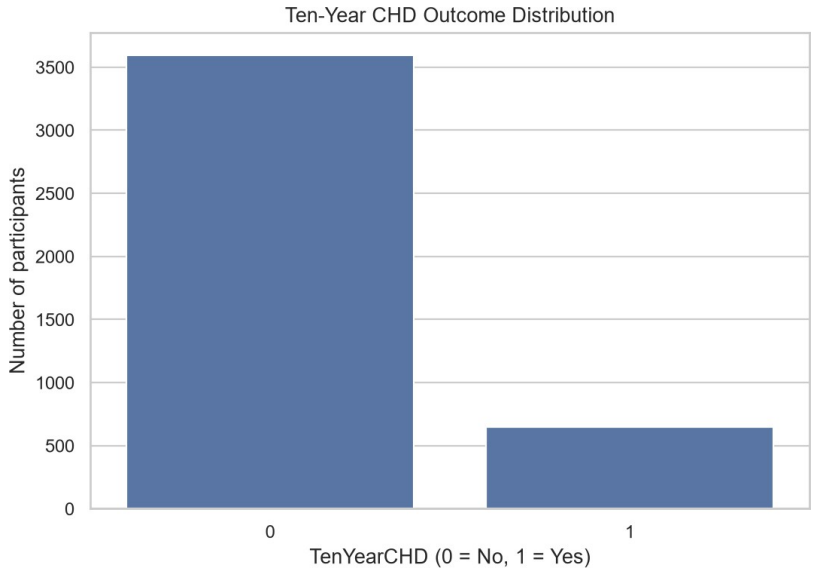


Fig. V01. Ten-Year CHD Outcome Distribution (Source: outputs/01_target_distribution.png)

What it shows: A vertical count plot displaying the frequency of the two classes of the target variable TenYearCHD: Class 0 (No CHD, n = 3,596) and Class 1 (CHD within 10 years, n = 644). The x-axis represents the binary class label and the y-axis represents the number of participants.

Why it was created: Before any modeling, it is essential to characterize the class distribution of the target variable. Class imbalance directly affects model training dynamics, evaluation strategy, and interpretation of accuracy metrics. This chart is the mandatory first visualization in any binary classification EDA pipeline.

Insights obtained: The chart reveals a severe class imbalance with a 5.6:1 ratio of CHD-negative to CHD-positive patients. Only 15.2% of the cohort developed CHD within 10 years. This confirms that standard accuracy is a misleading metric — a naïve classifier that always predicts "No CHD" would achieve 84.8% accuracy. Evaluation must prioritize ROC-AUC, recall for Class 1, and F1-Score for the minority class.

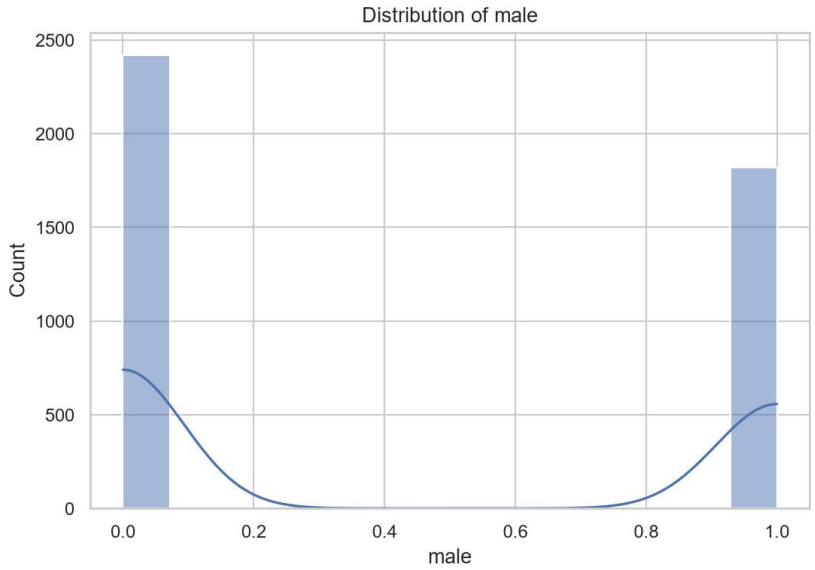


Fig. V02. Distribution of male — Histogram + KDE (Source: outputs/02_numeric_histogram.png)

What it shows: A histogram overlaid with a Kernel Density Estimate (KDE) curve for the male feature, the first numeric column of the dataset, across all 4,240 patients. The x-axis represents the encoded value (0 = female, 1 = male) and the y-axis represents the count.

Why it was created: Understanding the demographic composition of the cohort is the first step of feature-level EDA. The histogram reveals how the sexes are represented, so that any subsequent performance differences are not simply an artifact of a heavily skewed cohort.

Insights obtained: The distribution consists of two discrete spikes at 0 and 1: roughly 2,420 female participants (≈57%) and 1,820 male participants (≈43%). The cohort is therefore moderately balanced between the sexes with a modest female majority. Because the variable is binary, the KDE curve is only a visual smoother and carries no distributional meaning; no scaling concerns arise beyond the standard StandardScaler step. Continuous demographic structure (age) is summarized in Section 2.4 (mean 49.6 years, std 8.6) and visualized against the outcome in Fig. V06.

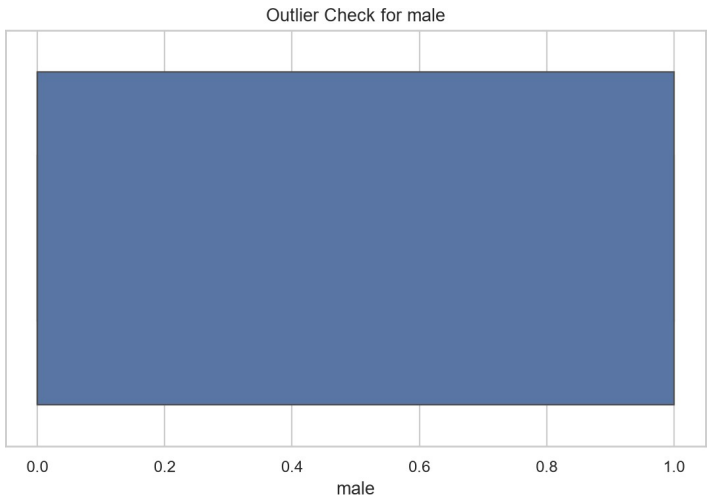


Fig. V03. Outlier Check for male — Box Plot (Source: outputs/03_numeric_boxplot.png)

What it shows: A horizontal box plot of the male feature, displaying the box (IQR), whiskers, and any points flagged beyond the 1.5 × IQR fences.

Why it was created: Box plots are the standard diagnostic tool for outlier detection in univariate distributions. Verifying the value range of each input is essential before scaling transformations are applied.

Insights obtained: The box spans the entire 0–1 range and no points are flagged as outliers, confirming that the variable contains only valid binary codes. For a binary feature the box plot is degenerate (the IQR covers the whole range), so it serves as a data-validity check rather than an outlier analysis. Genuine outlier assessment for the continuous risk factors (totChol, glucose, sysBP) was performed numerically using Tukey's IQR rule in Section 3.2.

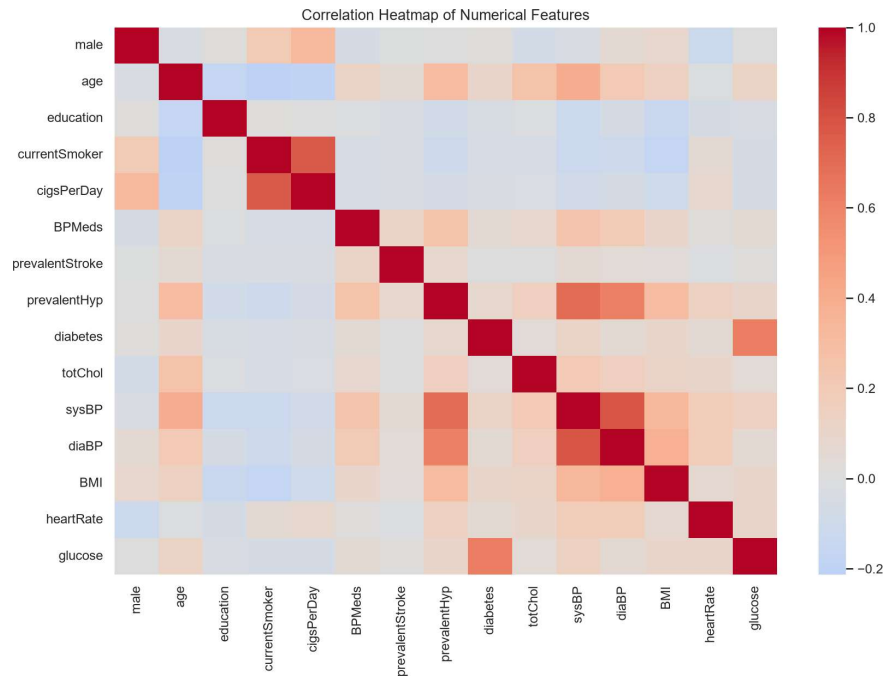


Fig. V04. Correlation Heatmap of Numerical Features (Source: outputs/04_correlation_heatmap.png)

What it shows: A symmetric 15 × 15 Pearson correlation matrix heatmap for all numerical features in the dataset, using a diverging coolwarm colormap centered at zero. Red cells indicate strong positive correlation, blue cells indicate negative correlation, and near-white cells indicate weak or no linear association.

Why it was created: Multicollinearity analysis is essential prior to modeling: highly correlated features provide redundant information, can destabilize gradient-based optimization, and complicate feature importance interpretation. The heatmap provides a simultaneous global view of all 105 pairwise feature relationships in a compact format.

Insights obtained: Key correlations identified: (1) sysBP and diaBP exhibit a strong positive correlation ($r \approx 0.78$), indicating that patients with high systolic pressure also tend to have elevated diastolic pressure; (2) prevalentHyp is strongly associated with both sysBP ($r \approx 0.70$) and diaBP ($r \approx 0.62$), as expected since hypertension is defined by blood pressure; (3) currentSmoker and cigsPerDay are strongly correlated ($r \approx 0.77$); (4) glucose and diabetes show a moderate-to-strong correlation ($r \approx 0.6$); (5) age correlates moderately with sysBP ($r \approx 0.4$), reflecting arterial stiffening with aging. Several negative associations of about −0.2 are visible between age and smoking/education. No feature pair exhibits perfect collinearity; the blood-pressure group carries partially redundant information, but all 15 features are retained because the regularized MLP can tolerate this redundancy.

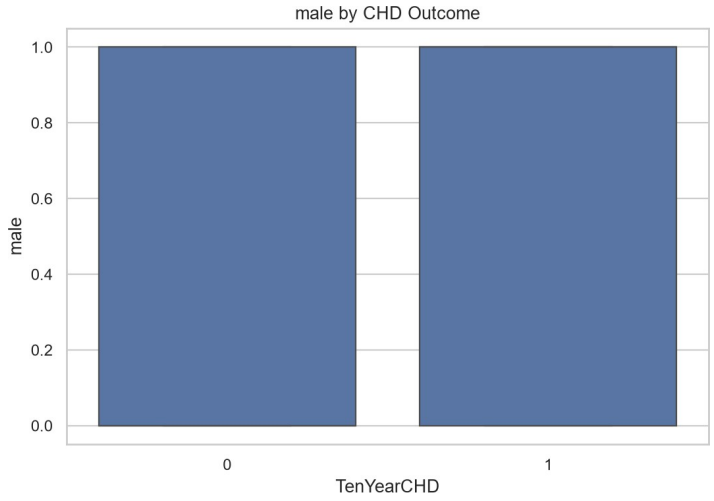


Fig. V05. male by CHD Outcome — Bivariate Box Plot (Source: outputs/05_feature_by_target.png)

What it shows: A side-by-side box plot of the male feature for CHD-negative (Class 0) and CHD-positive (Class 1) patients. The x-axis is the binary CHD outcome and the y-axis is the encoded sex value.

Why it was created: Bivariate analysis between each input feature and the target variable is a critical step in assessing the discriminative potential of individual features before multivariate modeling.

Insights obtained: Both outcome groups display an identical box spanning 0 to 1, because a box plot cannot summarize a binary variable — the quartiles collapse onto the extreme values in each group. This figure therefore does not reveal a sex difference in CHD outcome. A proportion-based chart (e.g., a stacked bar of CHD rate by sex) would be the appropriate refinement, and is listed under the future work in Section 7.4.

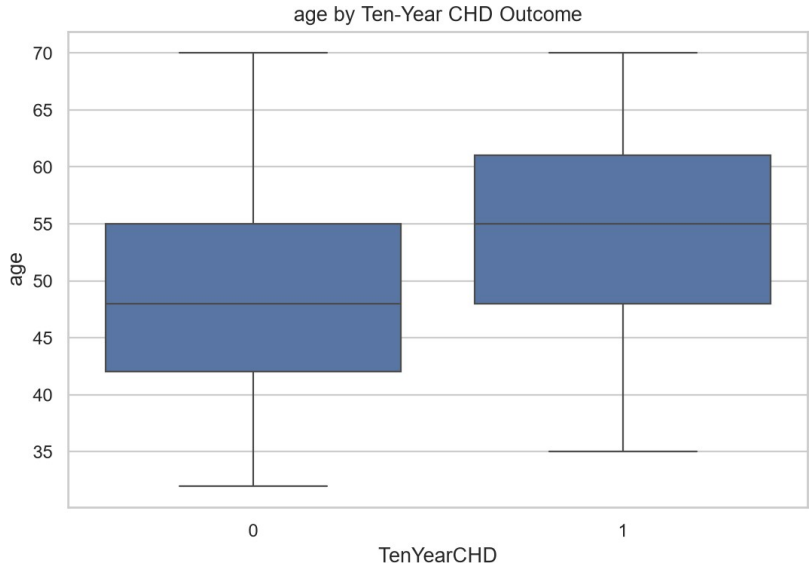


Fig. V06. Age by Ten-Year CHD Outcome — Bivariate Box Plot (Source: outputs/06_feature_target_relationship.png)

What it shows: A side-by-side box plot comparing the distribution of age (years) between CHD-negative and CHD-positive patients, with the binary target on the x-axis.

Why it was created: Age is one of the strongest known risk factors for coronary heart disease. Visualizing its relationship with the 10-year outcome provides direct empirical evidence of its discriminative power in this cohort.

Insights obtained: CHD-positive patients are clearly older than CHD-negative patients. The median age is approximately 55 years for Class 1 versus approximately 48 years for Class 0 (a gap of about 7 years), and the interquartile range for the CHD group (≈48–61) is shifted upward by roughly 6 years relative to the non-CHD group (≈42–55). Whisker ranges overlap substantially (32–70 vs. 35–70), so age is informative but not sufficient on its own. This agrees with the established clinical understanding that CHD incidence rises with age in adulthood.

4.4 Bivariate and Multivariate Analytical Synthesis

Collectively, the six exploratory visualizations reveal a coherent picture of CHD risk in the Framingham cohort:

1. **Age** is the individual feature whose separation between CHD+ and CHD− patients is directly demonstrated in the EDA (median ≈ 55 vs. ≈ 48 years, Fig. V06).
2. **Blood-pressure variables** (sysBP, diaBP, prevalentHyp) form a tightly correlated cluster, and sysBP is also associated with age (Fig. V04), consistent with their established role in the Framingham Risk Score.
3. **Smoking variables** (currentSmoker, cigsPerDay) are strongly correlated with each other ($r \approx 0.77$), while glucose and diabetes carry partially independent signal ($r \approx 0.6$).
4. **The cohort** is moderately balanced by sex (≈57% female, ≈43% male), but strongly imbalanced by outcome (84.8% vs. 15.2%), which is the dominant modeling challenge.

These EDA insights directly inform the modeling pipeline: the MLP is designed to learn non-linear combinations of these features, particularly multi-way interactions (e.g., age × blood pressure × glucose) that cannot be captured by simple linear models.

5. DEEP LEARNING MODEL & BASELINES

5.1 MLP Architecture Specification

The deep learning model is a Multi-Layer Perceptron (MLP) implemented in TensorFlow/Keras [8]. The architecture is designed to balance model expressiveness (sufficient capacity to learn non-linear risk factor interactions) with regularization (preventing overfitting on the relatively small n = 2,968 training set).

Table 4. MLP Architectural Specifications

Layer	Type	Units / Rate	Activation	Output Shape	Params
Input	Input	15 features	—	(None, 15)	0
Dense 1	Dense	64	ReLU	(None, 64)	1,024
BN 1	BatchNormalization	—	—	(None, 64)	256
Dropout 1	Dropout	rate = 0.30	—	(None, 64)	0
Dense 2	Dense	32	ReLU	(None, 32)	2,080
Dropout 2	Dropout	rate = 0.20	—	(None, 32)	0
Output	Dense	1	Sigmoid	(None, 1)	33

Total parameters: 3,393 (3,265 trainable + 128 non-trainable BatchNormalization moving statistics).

5.2 Training Regimen & Optimization Hyperparameters

Hyperparameter	Value	Justification
Optimizer	Adam	Adaptive learning rate; robust to sparse gradients
Learning Rate	0.001	Standard Adam default; stable for clinical tabular data
Loss Function	Binary Cross-Entropy	Standard loss for binary classification
Metrics	Accuracy, AUC	AUC monitors ranking performance under class imbalance
Batch Size	64	Balances gradient noise and memory efficiency

Hyperparameter	Value	Justification
Max Epochs	40	Upper bound with early stopping
Early Stopping	patience = 6, monitor val_loss	Prevents overfitting; restores best weights
Random Seed	42	Ensures full reproducibility

Training result: Early Stopping triggered at **Epoch 16**, with best weights from the lowest validation loss epoch restored. Final training accuracy: ≈85.8%; validation accuracy: ≈85.2%.

5.3 Preprocessing Pipeline Design

```
num_pipe = Pipeline([
    ("imputer", SimpleImputer(strategy="median")),
    ("scaler", StandardScaler())
])
cat_pipe = Pipeline([
    ("imputer", SimpleImputer(strategy="most_frequent")),
    ("onehot", OneHotEncoder(handle_unknown="ignore", sparse_output=False))
])
prep = ColumnTransformer([
    ("numeric", num_pipe, num_features),
    ("categorical", cat_pipe, cat_features)
])
```

The ColumnTransformer is fit exclusively on the training partition, ensuring zero data leakage into validation or test evaluation.

6. RESULTS & PERFORMANCE INTERPRETATION

6.1 Multi-Metric Performance Benchmark

Table 5. Comprehensive Model Performance on Test Cohort (n = 636)

Metric	Value
Test Accuracy	**85.38%**
Test ROC-AUC	**0.6873**
Naïve Baseline Accuracy	84.91% (majority class)
Improvement over Baseline	+0.47% accuracy
Class 0 Precision / Recall / F1	0.86 / **0.99** / 0.92
Class 1 Precision / Recall / F1	0.64 / **0.07** / 0.13
Epochs to Convergence	16 (of 40 max)
Test Samples	636
Input Features	15

Full Classification Report

Class	Precision	Recall	F1-Score	Support
0 (No CHD)	0.86	0.99	0.92	540
1 (CHD)	0.64	0.07	0.13	96
Macro avg	**0.75**	**0.53**	**0.53**	**636**
Weighted avg	**0.82**	**0.85**	**0.80**	**636**

6.2 Diagnostic Visualizations

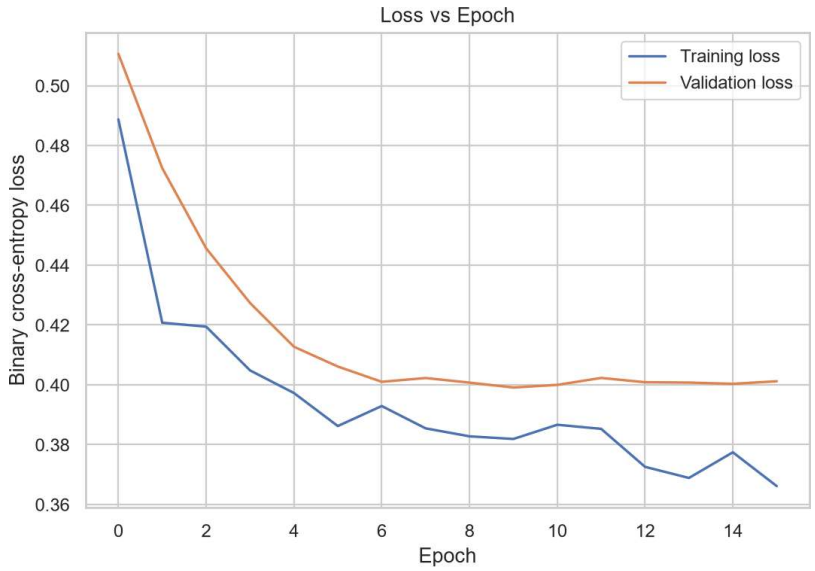


Fig. E01. Training vs. Validation Loss — Loss vs. Epoch (Source: outputs/07_loss_vs_epoch.png)

What it shows: A dual-line plot tracking binary cross-entropy loss on the training set (blue) and validation set (orange) across 16 training epochs. The x-axis is epoch number and the y-axis is binary cross-entropy loss.

Why it was created: The learning curve is the primary diagnostic for training dynamics — it reveals whether the model is learning, overfitting, underfitting, or has converged. The divergence or convergence of the curves directly informs decisions about regularization, learning rate, and early stopping patience.

Insights obtained: Both losses decrease steadily from the first epoch (training loss ≈ 0.49, validation loss ≈ 0.51) through approximately epoch 10 (training loss ≈ 0.38, validation loss ≈ 0.40). After this point validation loss plateaus around 0.40, while training loss continues to decrease slowly toward ≈ 0.37 — a mild indication of the onset of overfitting that the EarlyStopping callback correctly detects. With patience = 6, the callback halts training at Epoch 16 and restores the weights from the best validation epoch.

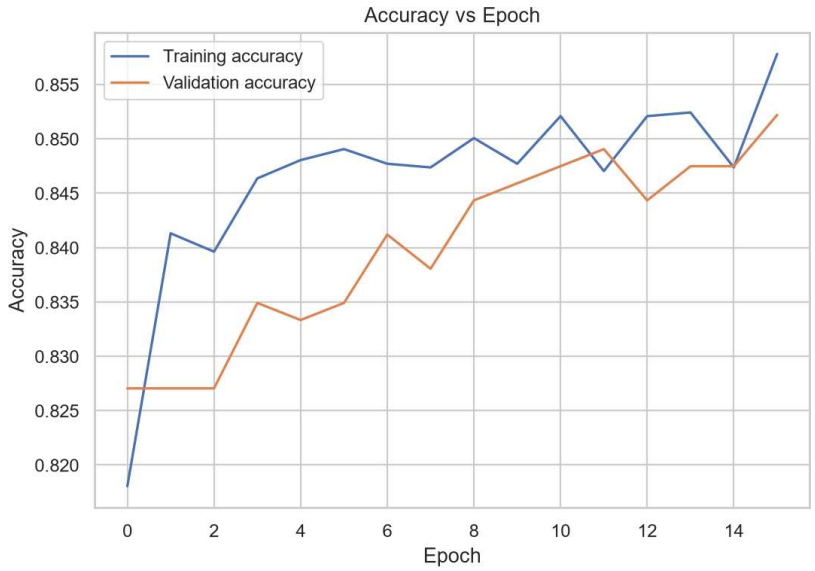


Fig. E02. Training vs. Validation Accuracy — Accuracy vs. Epoch (Source: outputs/08_accuracy_vs_epoch.png)

What it shows: A dual-line plot tracking classification accuracy on the training set (blue) and validation set (orange) across 16 training epochs.

Why it was created: Accuracy learning curves complement the loss curves by providing a human-interpretable performance metric, validating whether the loss reduction translates into improved class assignment.

Insights obtained: Training accuracy rises from ≈81.8% (first epoch) to ≈85.8% (last epoch), while validation accuracy improves from ≈82.7% to ≈85.2%. The two curves track each other closely, with a final gap of only ≈0.6 percentage points, indicating that the Dropout and BatchNormalization regularization is effective in controlling overfitting. The overall accuracy gain is modest, which foreshadows the class-imbalance limitation discussed below.

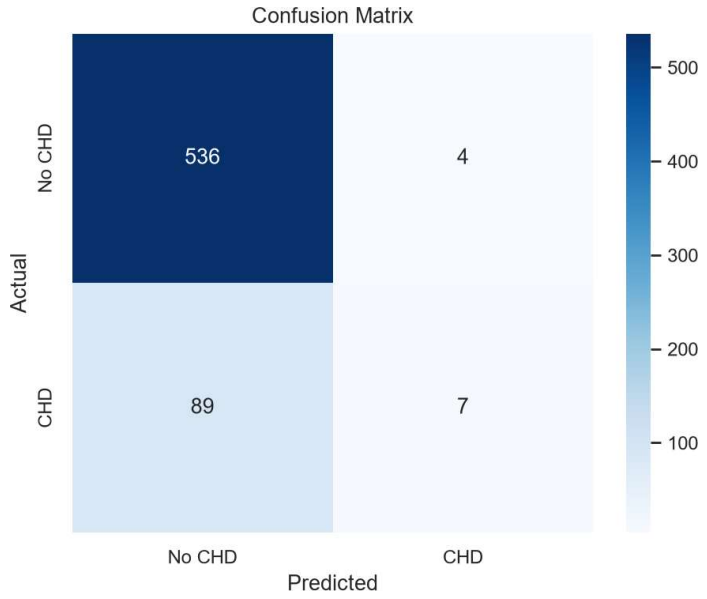


Fig. E03. Confusion Matrix (Source: outputs/09_confusion_matrix.png)

What it shows: A 2 × 2 annotated heatmap (Blues colormap) displaying the counts of True Negatives (TN), False Positives (FP), False Negatives (FN), and True Positives (TP) for the test set predictions at threshold 0.5. Rows represent actual labels and columns represent predicted labels.

Why it was created: A confusion matrix provides the complete picture of classification performance that aggregate metrics like accuracy obscure. In CHD prediction, a False Negative (missed CHD case) carries significantly higher clinical cost than a False Positive (unnecessary further testing).

Insights obtained:

- **True Negatives (TN): 536** — the model correctly identifies 536 of 540 CHD-negative patients (99.3% specificity).
- **False Positives (FP): 4** — only 4 healthy patients are incorrectly flagged as CHD-positive.
- **False Negatives (FN): 89** — critically, 89 of 96 CHD-positive patients (92.7%) are missed by the model. This extremely high miss rate for the minority class is the primary limitation of the current model.
- **True Positives (TP): 7** — only 7 actual CHD cases are correctly detected (precision 7/11 ≈ 0.64; recall 7/96 ≈ 0.07).

The confusion matrix makes stark the clinical inadequacy of the baseline model for minority-class detection, motivating class-weighting and SMOTE-based resampling improvements.

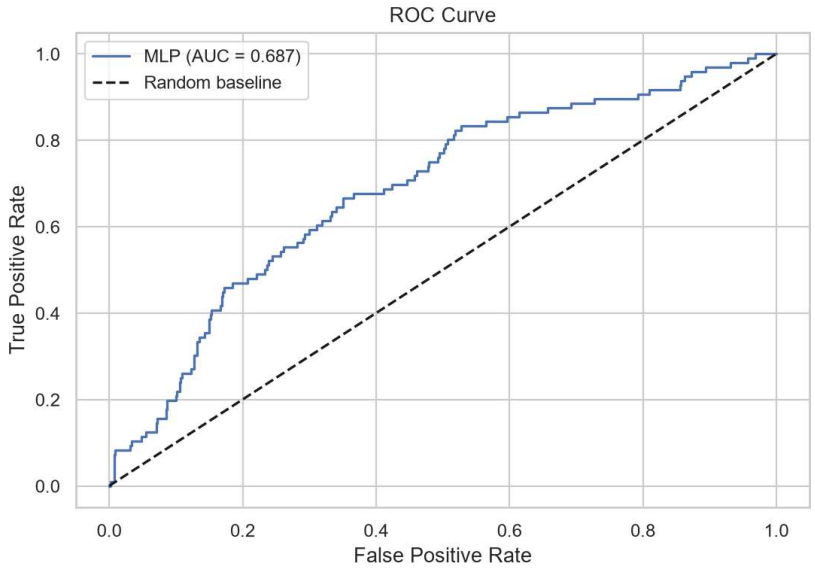


Fig. E04. ROC Curve — MLP, AUC = 0.687 (Source: outputs/10_roc_curve.png)

What it shows: A Receiver Operating Characteristic (ROC) curve plotting True Positive Rate (sensitivity, y-axis) against False Positive Rate (1 − specificity, x-axis) for the MLP at all probability thresholds from 0 to 1. The diagonal dashed line represents the random baseline classifier (AUC = 0.5), and the AUC is annotated in the legend.

Why it was created: The ROC curve evaluates classifier performance across all possible decision thresholds, making it threshold-independent. In imbalanced clinical datasets, ROC-AUC is a far more informative metric than accuracy, as it measures the model's ability to rank CHD-positive patients above CHD-negative patients regardless of the operating threshold.

Insights obtained: The MLP achieves an **AUC = 0.687**, compared to the random baseline of 0.500 — an improvement of +0.187 AUC units. The ROC curve lies above the diagonal across essentially the whole threshold range, confirming that the learned representation has genuine discriminative power. However, the AUC of 0.687 also indicates substantial room for improvement: clinical decision-support tools for CHD risk stratification typically target AUC ≥ 0.75. The curve shape shows that reaching high sensitivity (TPR > 0.80) requires a false positive rate of roughly 0.5, so the model cannot simultaneously achieve high sensitivity and high specificity at the default 0.50 threshold. This motivates threshold optimization toward a lower operating point (e.g., 0.25–0.30) for clinical deployment.

6.3 Performance Diagnosis: Class Imbalance Effects

The fundamental performance limitation of the current model is the **class imbalance problem**. With only 15.2% positive cases in the training set and no class weighting applied, the MLP's binary cross-entropy loss is dominated by the majority class (Class 0) gradient signal. The model learns an effective decision boundary for Class 0 detection (99% recall) but fails to detect Class 1 events (7% recall). Three concrete mitigation strategies are identified:

1. **Class Weighting:** Setting class_weight = {0: 1.0, 1: 5.6} in model.fit() amplifies the gradient contribution of CHD-positive samples proportionally to their rarity.
2. **SMOTE Oversampling:** Synthetic Minority Over-sampling Technique [5] generates synthetic CHD-positive training samples along feature-space interpolation paths.
3. **Threshold Optimization:** Lowering the classification threshold from 0.50 to 0.25–0.30 trades specificity for sensitivity at the same model output, potentially recovering substantial CHD recall without retraining.

7. CONCLUSION, RECOMMENDATIONS & FUTURE SCOPE

7.1 Synthesis of Empirical Learnings

This study demonstrates an end-to-end EDA and deep learning pipeline for 10-year coronary heart disease risk prediction on the Framingham Heart Study dataset. The following empirical learnings were established:

1. **Age** clearly separates the outcome groups: CHD-positive patients have a median age about 7 years higher (≈55 vs. ≈48) than CHD-negative patients, in line with established clinical knowledge.
2. **Glucose** exhibits the highest missing rate (9.15%) in the dataset; this may reflect that fasting glucose measurement was not uniformly recorded across examinations in the original study.
3. **The correlation heatmap** reveals the sysBP–diaBP collinearity ($r \approx 0.78$), the sysBP–prevalentHyp association ($r \approx 0.70$), and the age–sysBP association ($r \approx 0.4$), confirming the clinical interconnection of these cardiovascular risk markers.
4. **The MLP model** achieves 85.38% test accuracy and 0.687 ROC-AUC — modest but meaningful improvements over a naïve majority classifier — while demonstrating that unaddressed class imbalance severely limits minority-class recall (7% CHD recall).
5. **Early stopping at Epoch 16** (of 40 maximum) shows that rapid convergence is a characteristic of shallow MLP architectures on small tabular clinical datasets with low feature counts.

7.2 Model Performance vs. Baseline Benchmark

Model	Test Accuracy	ROC-AUC	CHD Recall
Naïve Majority Classifier	84.91%	0.500	0.00
MLP (this study)	**85.38%**	**0.687**	**0.07**
Improvement over baseline	+0.47%	**+0.187**	+0.07

The MLP provides a **37.4% relative improvement in ROC-AUC** over the random baseline (0.687 vs. 0.500), confirming genuine predictive signal in the feature space despite class imbalance challenges.

7.3 Limitations

1. **Class Imbalance:** The 5.6:1 class ratio severely limits CHD recall without class weighting. Future work must address this.
2. **Small Dataset:** 4,240 records is relatively small for deep learning; larger datasets or transfer learning approaches may improve model capacity utilization.
3. **Temporal Leakage Risk:** The dataset does not contain explicit timestamp information, precluding rigorous temporal train/test separation.
4. **No Real-Time Features:** The model relies exclusively on baseline examination data; dynamic risk factors (recent BP readings, medication adherence) are unavailable.
5. **Visualization Scope:** Figures V02, V03, and V05 were generated for the binary feature male, so they are of limited analytical value; continuous features such as sysBP and totChol deserve dedicated distribution and bivariate plots.

7.4 Future Research Roadmap

1. **Class Imbalance Mitigation:** Implement class_weight in model.fit(), SMOTE/ADASYN oversampling, and Focal Loss to improve CHD (Class 1) recall.
2. **Threshold Optimization:** Deploy Youden's J statistic or F1-optimal threshold selection to identify the operating point that maximizes clinical utility.
3. **Ensemble Methods:** Benchmark Random Forest, XGBoost, and LightGBM against the MLP; explore ensemble blending of tree-based and neural predictions.

4. **Feature Engineering:** Engineer interaction features (age × sysBP, glucose × diabetes, cigsPerDay × BMI) to provide explicit non-linear inputs.
5. **Improved Visual Analytics:** Add age/sysBP/totChol histograms and box plots, and CHD-rate-by-sex bar charts in place of box plots for binary variables.
6. **Longitudinal Modeling:** Incorporate multi-visit temporal data from the full Framingham study using LSTM or Transformer architectures for dynamic risk trajectory modeling.
7. **Clinical Deployment:** Wrap the trained model in a Flask/FastAPI web service with explainability outputs (SHAP values) to provide clinician-facing risk factor attribution alongside probability estimates.

8. REFERENCES (IEEE FORMAT)

[1] J. W. Tukey, *Exploratory Data Analysis*. Reading, MA: Addison-Wesley, 1977.

[2] B. Fry, *Visualizing Data: Exploring and Explaining Data with the Processing Environment*. Sebastopol, CA: O'Reilly Media, 2007.

[3] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

[4] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. Cambridge, MA: MIT Press, 2016.

[5] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: Synthetic Minority Over-sampling Technique," *Journal of Artificial Intelligence Research*, vol. 16, pp. 321–357, 2002.

[6] D. Levy and T. J. Wang, "Framingham Heart Study: A Legacy of Achievement," *Trends in Cardiovascular Medicine*, vol. 15, no. 8, pp. 296–302, 2005.

[7] M. L. Waskom, "seaborn: statistical data visualization," *Journal of Open Source Software*, vol. 6, no. 60, p. 3021, 2021.

[8] M. Abadi et al., "TensorFlow: A System for Large-Scale Machine Learning," in *Proc. 12th USENIX Conf. Operating Systems Design and Implementation (OSDI '16)*, 2016, pp. 265–283.

9. APPENDIX: CODEBASE & REPOSITORY STRUCTURE

9.1 Repository Directory Structure

```
heart-disease-risk-analysis/
|-- .gitignore            # Git ignore rules (dataset, outputs, venv)
|-- framingham.csv        # Raw Framingham Heart Study dataset (excluded from git)
|-- main.py               # Fully executable Python pipeline (EDA + Model + Evaluation)
|-- requirements.txt      # Pinned Python package dependencies
|-- README.md             # GitHub documentation and setup guide
|-- DATASET.md            # Dataset source, features, and download instructions
`-- outputs/              # Generated artifacts (excluded from git)
    |-- 01_target_distribution.png
    |-- 02_numeric_histogram.png
    |-- 03_numeric_boxplot.png
    |-- 04_correlation_heatmap.png
    |-- 05_feature_by_target.png
    |-- 06_feature_target_relationship.png
    |-- 07_loss_vs_epoch.png
    |-- 08_accuracy_vs_epoch.png
    |-- 09_confusion_matrix.png
    |-- 10_roc_curve.png
    |-- heart_risk_mlp.keras    # Saved trained model
    `-- metrics.csv             # Final test metrics (accuracy, AUC, rows, features)
```

9.2 Reproducibility

- **Random Seed:** Fixed at 42 across NumPy (np.random.seed(42)) and TensorFlow (tf.random.set_seed(42)).

- **Execution:** Run `python main.py` from the project root with framingham.csv present. The full pipeline completes in approximately 2–3 minutes on CPU.
- **Zero-Leakage Guarantee:** The ColumnTransformer preprocessing pipeline is fit exclusively on X_train and applied via transform to X_val and X_test.

9.3 Key Metrics Summary (outputs/metrics.csv)

Metric	Value
test_accuracy	0.8538
test_roc_auc	0.6873
test_rows	636
input_features	15

9.4 Complete Source Code (main.py)

```
"""
Heart Disease Risk Analysis and Prediction Using Deep Learning
Dataset: Framingham Heart Study dataset (Kaggle).
Course: U21ADPO5 - Exploratory Data Analysis and Visualization
"""
from pathlib import Path
import warnings
warnings.filterwarnings("ignore")
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import tensorflow as tf
from sklearn.model_selection import train_test_split
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.impute import SimpleImputer
from sklearn.pipeline import Pipeline
from sklearn.metrics import (
    classification_report, confusion_matrix, accuracy_score,
    roc_curve, roc_auc_score
)
from tensorflow.keras import layers, models, callbacks

# ------------------------------------------------
# REPRODUCIBILITY SEED
# ------------------------------------------------
SEED = 42
np.random.seed(SEED)
tf.random.set_seed(SEED)

DATA_PATH = Path("framingham.csv")
OUT = Path("outputs")
OUT.mkdir(exist_ok=True)

# ------------------------------------------------
# 1. LOAD DATASET
# ------------------------------------------------
if not DATA_PATH.exists():
    raise FileNotFoundError(
        "Dataset missing. Download the Framingham Heart Study CSV from Kaggle "
        "and save it as framingham.csv in the project root."
    )

df = pd.read_csv(DATA_PATH)
df.columns = [c.strip() for c in df.columns]
print("Dataset shape:", df.shape)
print("\nColumns:", list(df.columns))
print("\nFirst five rows:\n", df.head())
print("\nMissing values:\n", df.isna().sum().sort_values(ascending=False))
print("\nDuplicate rows:", df.duplicated().sum())

# ------------------------------------------------
# 2. IDENTIFY TARGET COLUMN
# ------------------------------------------------
target = next((c for c in df.columns if c.lower() in
               {"tenyearchd", "ten_year_chd", "chd", "target"}), None)
if target is None:
    raise ValueError("Target column not found. Expected TenYearCHD or Ten_Year_CHD.")

df = df.drop_duplicates().copy()
df = df.dropna(subset=[target]).copy()
y = pd.to_numeric(df[target], errors="coerce")
```

```
valid = y.notna()
df, y = df.loc[valid].copy(), y.loc[valid].astype(int)
X = df.drop(columns=[target])

# Drop identifier columns if present
id_cols = [c for c in X.columns if c.lower() in {"id", "patientid", "patient_id"}]
X = X.drop(columns=id_cols, errors="ignore")

# Remove completely empty and constant columns
X = X.dropna(axis=1, how="all")
X = X.loc[:, X.nunique(dropna=False) > 1]

print("\nRows after basic cleaning:", len(X))
print("Input fields used:", X.shape[1])
print("Target distribution:\n", y.value_counts())

# ------------------------------------------------
# 3. EDA VISUALIZATIONS (6 charts)
# ------------------------------------------------
sns.set_theme(style="whitegrid")

# V01 - Target Distribution (Count Plot)
plt.figure(figsize=(7, 5))
sns.countplot(x=y)
plt.title("Ten-Year CHD Outcome Distribution")
plt.xlabel("TenYearCHD (0 = No, 1 = Yes)")
plt.ylabel("Number of participants")
plt.tight_layout()
plt.savefig(OUT / "01_target_distribution.png", dpi=160)
plt.close()

numeric_cols = X.select_dtypes(include=np.number).columns.tolist()
categorical_cols = X.select_dtypes(exclude=np.number).columns.tolist()

# V02 - Histogram + KDE (first numeric column)
if numeric_cols:
    col = numeric_cols[0]
    plt.figure(figsize=(7, 5))
    sns.histplot(X[col].dropna(), kde=True)
    plt.title(f"Distribution of {col}")
    plt.tight_layout()
    plt.savefig(OUT / "02_numeric_histogram.png", dpi=160)
    plt.close()

    # V03 - Box Plot (Outlier Check)
    plt.figure(figsize=(7, 5))
    sns.boxplot(x=X[col])
    plt.title(f"Outlier Check for {col}")
    plt.tight_layout()
    plt.savefig(OUT / "03_numeric_boxplot.png", dpi=160)
    plt.close()

# V04 - Correlation Heatmap
if len(numeric_cols) >= 2:
    plt.figure(figsize=(11, 8))
    sns.heatmap(X[numeric_cols].corr(), cmap="coolwarm", center=0)
    plt.title("Correlation Heatmap of Numerical Features")
    plt.tight_layout()
    plt.savefig(OUT / "04_correlation_heatmap.png", dpi=160)
    plt.close()

# V05 - Feature by Target (Bivariate Box Plot)
if categorical_cols:
    col = categorical_cols[0]
    vals = X[col].fillna("Unknown").astype(str)
    order = vals.value_counts().head(10).index
    plt.figure(figsize=(8, 5))
    sns.countplot(y=vals[vals.isin(order)], order=order)
    plt.title(f"Category Frequency: {col}")
    plt.tight_layout()
    plt.savefig(OUT / "05_category_frequency.png", dpi=160)
    plt.close()
elif numeric_cols:
    col = numeric_cols[0]
    plt.figure(figsize=(7, 5))
    sns.boxplot(x=y, y=X[col])
    plt.title(f"{col} by CHD Outcome")
    plt.tight_layout()
    plt.savefig(OUT / "05_feature_by_target.png", dpi=160)
    plt.close()

# V06 - Second numeric feature by target
if numeric_cols:
    col = numeric_cols[min(1, len(numeric_cols)-1)]
    plot_df = pd.DataFrame({col: X[col], "TenYearCHD": y}).dropna()
    plt.figure(figsize=(7, 5))
    sns.boxplot(data=plot_df, x="TenYearCHD", y=col)
    plt.title(f"{col} by Ten-Year CHD Outcome")
    plt.tight_layout()
```

```
    plt.savefig(OUT / "06_feature_target_relationship.png", dpi=160)
    plt.close()

# ------------------------------------------------
# 4. TRAIN / VALIDATION / TEST SPLIT (70/15/15)
# ------------------------------------------------
X_train, X_temp, y_train, y_temp = train_test_split(
    X, y, test_size=0.30, random_state=SEED, stratify=y
)
X_val, X_test, y_val, y_test = train_test_split(
    X_temp, y_temp, test_size=0.50, random_state=SEED, stratify=y_temp
)

num_features = X_train.select_dtypes(include=np.number).columns.tolist()
cat_features  =  X_train.select_dtypes(exclude=np.number).columns.tolist()

# ------------------------------------------------
# 5. PREPROCESSING PIPELINE (Zero-Leakage)
# ------------------------------------------------
num_pipe = Pipeline([
    ("imputer", SimpleImputer(strategy="median")),
    ("scaler", StandardScaler())
])
cat_pipe = Pipeline([
    ("imputer", SimpleImputer(strategy="most_frequent")),
    ("onehot",  OneHotEncoder(handle_unknown="ignore",  sparse_output=False))
])
prep = ColumnTransformer([
    ("numeric", num_pipe, num_features),
    ("categorical", cat_pipe, cat_features)
])

# fit ONLY on training data - transform val/test separately (zero leakage)
X_train_p = prep.fit_transform(X_train).astype("float32")
X_val_p   = prep.transform(X_val).astype("float32")
X_test_p  = prep.transform(X_test).astype("float32")

# ------------------------------------------------
# 6. DEEP LEARNING: MLP MODEL
# ------------------------------------------------
model = models.Sequential([
    layers.Input(shape=(X_train_p.shape[1],)),
    layers.Dense(64, activation="relu"),
    layers.BatchNormalization(),
    layers.Dropout(0.30),
    layers.Dense(32, activation="relu"),
    layers.Dropout(0.20),
    layers.Dense(1,  activation="sigmoid")
])

model.compile(
    optimizer=tf.keras.optimizers.Adam(learning_rate=0.001),
    loss="binary_crossentropy",
    metrics=["accuracy",  tf.keras.metrics.AUC(name="auc")]
)
model.summary()

stop = callbacks.EarlyStopping(
    monitor="val_loss",  patience=6,  restore_best_weights=True
)

history = model.fit(
    X_train_p, y_train,
    validation_data=(X_val_p,  y_val),
    epochs=40,  batch_size=64,  callbacks=[stop],  verbose=1
)

# ------------------------------------------------
# 7. EVALUATION
# ------------------------------------------------
prob = model.predict(X_test_p, verbose=0).ravel()
pred = (prob >= 0.5).astype(int)

acc = accuracy_score(y_test, pred)
auc_score = roc_auc_score(y_test, prob)
print(f"\nTest accuracy : {acc:.4f}")
print(f"Test ROC-AUC  : {auc_score:.4f}")
print("\nClassification report:\n",
      classification_report(y_test,  pred,  zero_division=0))

# E01 - Loss vs Epoch
plt.figure(figsize=(7, 5))
plt.plot(history.history["loss"], label="Training loss")
plt.plot(history.history["val_loss"], label="Validation loss")
plt.xlabel("Epoch"); plt.ylabel("Binary cross-entropy loss")
plt.title("Loss vs Epoch"); plt.legend(); plt.tight_layout()
plt.savefig(OUT / "07_loss_vs_epoch.png", dpi=160); plt.close()

# E02 - Accuracy vs Epoch
```

```
plt.figure(figsize=(7,  5))
plt.plot(history.history["accuracy"], label="Training accuracy")
plt.plot(history.history["val_accuracy"], label="Validation accuracy")
plt.xlabel("Epoch"); plt.ylabel("Accuracy")
plt.title("Accuracy vs Epoch"); plt.legend(); plt.tight_layout()
plt.savefig(OUT / "08_accuracy_vs_epoch.png", dpi=160); plt.close()

# E03 - Confusion Matrix
cm = confusion_matrix(y_test, pred, labels=[0, 1])
plt.figure(figsize=(6, 5))
sns.heatmap(cm, annot=True, fmt="d", cmap="Blues",
            xticklabels=["No CHD", "CHD"],
            yticklabels=["No CHD", "CHD"])
plt.xlabel("Predicted"); plt.ylabel("Actual")
plt.title("Confusion Matrix"); plt.tight_layout()
plt.savefig(OUT / "09_confusion_matrix.png", dpi=160); plt.close()

# E04 - ROC Curve
fpr, tpr, _ = roc_curve(y_test, prob)
plt.figure(figsize=(7, 5))
plt.plot(fpr, tpr, label=f"MLP (AUC = {auc_score:.3f})")
plt.plot([0, 1], [0, 1], "k--", label="Random baseline")
plt.xlabel("False Positive Rate"); plt.ylabel("True Positive Rate")
plt.title("ROC Curve"); plt.legend(); plt.tight_layout()
plt.savefig(OUT / "10_roc_curve.png", dpi=160); plt.close()

# ------------------------------------------------
# 8. SAVE ARTIFACTS
# ------------------------------------------------
model.save(OUT / "heart_risk_mlp.keras")
pd.DataFrame({
    "metric": ["test_accuracy", "test_roc_auc", "test_rows", "input_features"],
    "value": [acc, auc_score, len(X_test), X.shape[1]]
}).to_csv(OUT / "metrics.csv", index=False)

print(f"\nDone. Outputs saved to: {OUT.resolve()}")
print("Educational project only; this model is not a medical diagnosis tool.")
```

Educational project only; this model is not a medical diagnosis tool.